Week6_practice_arXiv

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In this case study, I use the API provided by arXiv to retrieve research paper metadata (arXiv 2026). I used the arXiv API to retrieve up to 500 research papers matching the exact phrase "climate change" through the all search field.

```
#install.packages("httr")  # Run only once
library(httr)
library(xml2)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.4     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
arxiv <- GET("http://export.arxiv.org/api/query?search_query=all:climate%20change")

status_code(arxiv)
```

```
[1] 200
```

```
content(arxiv) |>
  read_xml() |>
  html_structure()
```

```
<feed [xmlns:opensearch, xmlns:arxiv, xmlns]>
  <id>
    {text}
  <title>
    {text}
  <updated>
    {text}
  <link [href, type]>
  <itemsPerPage>
    {text}
  <totalResults>
    {text}
  <startIndex>
    {text}
  <entry>
    <id>
      {text}
    <title>
      {text}
    <updated>
      {text}
    <link [href, rel, type]>
    <link [href, rel, type, title]>
    <summary>
      {text}
    <category [term, scheme]>
    <published>
      {text}
    <comment>
      {text}
    <primary_category [term]>
    <journal_ref>
      {text}
    <author>
      <name>
        {text}
    <doi>
      {text}
    <link [rel, href, title]>
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    <id>
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  <summary>
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  <title>
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  <updated>
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  <link [href, rel, type]>
  <link [href, rel, type, title]>
  <summary>
    {text}
```

```
    <category [term, scheme]>
    <category [term, scheme]>
    <published>
      {text}
    <primary_category [term]>
    <journal_ref>
      {text}
    <author>
      <name>
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    <author>
      <name>
        {text}
    <author>
      <name>
        {text}
    <doi>
      {text}
    <link [rel, href, title]>
```

```
content(arxiv) |>
  read_xml()
```

```
{xml_document}
<feed xmlns:opensearch="http://a9.com/-/spec/opensearch/1.1/" xmlns:arxiv="http://arxiv.org/s
 [1] <id>https://arxiv.org/api/KmRq2mBvH0RiZcZkTKGayGbRctM</id>
 [2] <title>arXiv Query: search_query=all:climate OR all:change&amp;id_list=& ...
 [3] <updated>2026-10-19T00:40:41Z</updated>
 [4] <link href="https://arxiv.org/api/query?search_query=all:climate+OR+all: ...
 [5] <opensearch:itemsPerPage>10</opensearch:itemsPerPage>
 [6] <opensearch:totalResults>200405</opensearch:totalResults>
 [7] <opensearch:startIndex>0</opensearch:startIndex>
 [8] <entry>\n  <id>http://arxiv.org/abs/1503.07557v1</id>\n  <title>Precipit ...
 [9] <entry>\n  <id>http://arxiv.org/abs/2012.10386v1</id>\n  <title>Future C ...
[10] <entry>\n  <id>http://arxiv.org/abs/2008.01035v1</id>\n  <title>Invasive ...
[11] <entry>\n  <id>http://arxiv.org/abs/2409.18574v2</id>\n  <title>Climate  ...
[12] <entry>\n  <id>http://arxiv.org/abs/1704.00563v1</id>\n  <title>Impact o ...
[13] <entry>\n  <id>http://arxiv.org/abs/2012.00483v2</id>\n  <title>ClimaTex ...
[14] <entry>\n  <id>http://arxiv.org/abs/2011.12443v1</id>\n  <title>The Huma ...
[15] <entry>\n  <id>http://arxiv.org/abs/2007.00239v1</id>\n  <title>Zonally  ...
[16] <entry>\n  <id>http://arxiv.org/abs/2305.00975v1</id>\n  <title>Deep Ens ...
[17] <entry>\n  <id>http://arxiv.org/abs/2504.20056v1</id>\n  <title>Fragilit ...
```

```
content(arxiv) |>
  read_xml() |>
  xml_child(search = 8)|>
  xml_child(search = 2)
```

```
{xml_node}
<title>
```

```
content(arxiv) |>
  read_xml() |>
  xml_child(search = 8)|>
  xml_child(search = 2) |>
  xml_text()
```

```
[1] "Precipitation extremes under climate change"
```

```
xml_data <- read_xml(arxiv)
xml_ns_strip(xml_data)
titles <- xml_text(xml_find_all(xml_data, ".//title"))
titles
```

```
 [1] "arXiv Query: search_query=all:climate OR all:change&id_list=&start=0&max_results=10"
 [2] "Precipitation extremes under climate change"
 [3] "Future Climate Change Projections over the Indian Region"
 [4] "Invasive species, extreme fire risk, and toxin release under a changing climate"
 [5] "Climate Adaptation with Reinforcement Learning: Experiments with Flooding and Transport
 [6] "Impact of anthropogenic climate change on the East Asian summer monsoon"
 [7] "ClimaText: A Dataset for Climate Change Topic Detection"
 [8] "The Human Effect Requires Affect: Addressing Social-Psychological Factors of Climate Ch
 [9] "Zonally opposing shifts of the intertropical convergence zone in response to climate ch
[10] "Deep Ensembles to Improve Uncertainty Quantification of Statistical Downscaling Models
[11] "Fragility Modeling of Power Grid Infrastructure for Addressing Climate Change Risks and
```

```
dates <- xml_text(xml_find_all(xml_data, ".//published"))
dates
```

```
 [1] "2015-03-25T21:11:05Z" "2020-12-02T07:46:57Z" "2020-08-03T17:22:18Z"
 [4] "2024-09-27T09:18:57Z" "2017-04-03T13:12:26Z" "2020-12-01T13:42:37Z"
 [7] "2020-11-24T23:34:54Z" "2020-07-01T04:48:26Z" "2023-04-27T19:53:18Z"
[10] "2025-04-11T10:20:47Z"
```

```r
df <- data.frame(titles = titles[-1], dates = dates)
df
```

```
1                                                                          Precipitation e
2                                                                Future Climate Change Projec
3                                                  Invasive species, extreme fire risk, and toxin rele
4              Climate Adaptation with Reinforcement Learning: Experiments with Flooding and
5                                                  Impact of anthropogenic climate change on t
6                                                                  ClimaText: A Dataset for Cl
7       The Human Effect Requires Affect: Addressing Social-Psychological Factors of Climate
8                             Zonally opposing shifts of the intertropical convergence zone i
9  Deep Ensembles to Improve Uncertainty Quantification of Statistical Downscaling Models und
10                    Fragility Modeling of Power Grid Infrastructure for Addressing Climate
                   dates
1  2015-03-25T21:11:05Z
2  2020-12-02T07:46:57Z
3  2020-08-03T17:22:18Z
4  2024-09-27T09:18:57Z
5  2017-04-03T13:12:26Z
6  2020-12-01T13:42:37Z
7  2020-11-24T23:34:54Z
8  2020-07-01T04:48:26Z
9  2023-04-27T19:53:18Z
10 2025-04-11T10:20:47Z
```

```r
library(httr2)
```

```
Attaching package: 'httr2'

The following object is masked from 'package:xml2':

    url_parse
```

```r
library(xml2)
library(tidyverse)

# --- Configuration ---
query_term    <- 'all:"climate change"'  # Change to your target search query
```

```r
batch_size    <- 100            # Max allowed per request by arXiv is 30,000
total_results <- 500            # Total number of papers you want to fetch
base_url      <- "http://export.arxiv.org/api/query"

# Empty list container to store tibbles from each batch iteration
all_batches <- list()

# --- Loop Through Results ---
for (start_idx in seq(0, total_results - 1, by = batch_size)) {

  message(sprintf("Fetching results %d to %d...", start_idx, start_idx + batch_size - 1))

  # 1. Build and execute the API request
  req <- request(base_url) %>%
    req_url_query(
      search_query = query_term,
      start = start_idx,
      max_results = batch_size
    )

  resp <- req_perform(req)
  xml_content <- resp_body_xml(resp)

  # 2. Extract elements using XML namespaces (arXiv uses the Atom 1.0 standard)
  # Strip namespaces to make standard XPath selectors easier
  xml_ns_strip(xml_content)

  entries <- xml_find_all(xml_content, ".//entry")

  # Break loop early if the API returns no more records
  if (length(entries) == 0) {
    message("No more entries found. Ending loop.")
    break
  }

  # 3. Parse Title and Publication Date from the current batch
  titles       <- xml_text(xml_find_all(entries, "./title"))
  pub_dates    <- xml_text(xml_find_all(entries, "./published"))

  # Clean up formatting strings (removes line breaks/extra spaces from titles)
  titles       <- str_squish(titles)
  pub_dates    <- as.Date(pub_dates) # Converts "2026-10-12T14:00:00Z" to date class
```

```
  # 4. Save current batch into our list container
  all_batches[[length(all_batches) + 1]] <- tibble(
    title = titles,
    publication_date = pub_dates
  )

  # 5. Respect the API Rules: arXiv requires a 3-second delay between calls
  Sys.sleep(3)
}
```

```
Fetching results 0 to 99...

Fetching results 100 to 199...

Fetching results 200 to 299...

Fetching results 300 to 399...

Fetching results 400 to 499...
```

```
# --- Combine Into One Large Tibble ---
big_arxiv_tibble <- bind_rows(all_batches)

# View the final dataset
print(big_arxiv_tibble)
```

```
# A tibble: 500 x 2
   title                                                  publication_date
   <chr>                                                  <date>
 1 Precipitation extremes under climate change            2015-03-25
 2 Future Climate Change Projections over the Indian Region 2020-12-02
 3 ClimaText: A Dataset for Climate Change Topic Detection 2020-12-01
 4 The Human Effect Requires Affect: Addressing Social-Psychol~ 2020-11-24
 5 Deep Ensembles to Improve Uncertainty Quantification of Sta~ 2023-04-27
 6 Fragility Modeling of Power Grid Infrastructure for Address~ 2025-04-11
 7 Learning Radiative Transfer Models for Climate Change Appli~ 2019-06-08
 8 An Interpretable Model of Climate Change Using Correlative ~ 2022-12-05
 9 Southern Ocean Dynamics Under Climate Change: New Knowledge~ 2023-10-21
10 Establishing an Evaluation Metric to Quantify Climate Chang~ 2019-10-22
# i 490 more rows
```

arXiv. 2026. “arXiv API User’s Manual.” https://info.arxiv.org/help/api/index.html.